MATH317 · NUMERICAL METHODS

Homework 2 Practice

Problems 2.1 · Floating-point

Muharib

01 Questions

Answer each question in the space below it. The full solution is in a separate file.

4(f) SINGLE + DOUBLE

Determine the single-precision and double-precision machine representation of 64.37109375.

4(c) SINGLE + DOUBLE

Determine the single-precision and double-precision machine representation of −9876.54321.

8(c) HEX TO DECIMAL

Determine the decimal number that has the machine representation $[494F96A0]_{16}$.

11 SMALL SYSTEM

Enumerate the set of numbers in the floating-point number system that have binary representations of the form $\pm(0.b_1b_2)_2 \times 2^k$, where $k \in \{-1, 0\}$.

15 UNDERFLOW

In the case of machine underflow, what is the relative error involved in replacing a number x by zero?

18 ROUNDING

In a five-decimal machine that correctly rounds numbers to the nearest machine number, what real numbers x will have the property fl(1.0 + x) = 1.0?

21 ROUNDOFF ERROR

(Continuation) What is the relative roundoff error when we round off $2^{-1} + 2^{-26}$ to get the closest machine number?

38 LOSS OF SIGNIFICANCE

Consider a computer that uses five-decimal-digit numbers. Show that if $x = 0.5321487513$ and $y = 0.5321304421$, then the operation $fl(x) - fl(y)$ involves a large relative error. Compute it.